

BalaSense

Power management & battery strategy

Technical document — May 2026 · Hardware: ESP32 Dev Module + 6 sensors

1. Context & constraints

BalaSense is a real-time physiological monitoring system embedded in a racing driver's balaclava. The target context is a local race stand with a car equipped with standard 12V power. This single constraint shapes the entire power strategy.

1.1 System power budget

Estimated consumption in continuous operation:

Component	Consumption	Notes
ESP32 (WiFi active)	160–240 mA	Peak 340 mA on TX burst
ICM-20948 (IMU)	~3 mA	Normal mode, 6 axes
SCD41 (CO ₂)	~18 mA	Measurement every 5 s
SDP810 (resp. flow)	~3 mA	Continuous measurement
TMP117 (skin temp)	~1.5 mA	Conversion every 1 s
MAX30003 (ECG)	~3 mA	AFE + FIFO active
CJMCU-6701 (GSR)	~2 mA	Continuous analog
3.3V regulator	~5 mA	Static losses
TOTAL	~300–400 mA	Continuous operation

NOTE At 400 mA average draw, the 1200 mAh LiPo provides ~3h of continuous operation — sufficient for all race session types. With firmware sleep optimisation (target <150 mA average), autonomy reaches ~8h.

1.2 Motorsport-specific constraints

- Space: the balaclava sits directly against skin under the helmet. All added components must be flat and lightweight.
- Heat: inside a racing helmet can reach 40–50°C, directly affecting battery performance and safety.
- Session length: from 20 min (sprint karting) to 3 h (endurance). The strategy must cover the longest case.
- FIA regulation: electronic devices and batteries near the cockpit are subject to safety requirements (fire-resistant materials, short-circuit protection).

2. Three power architectures evaluated

Option A — Wired from the car

A discrete cable from the dashboard (12V or 5V via DC-DC regulator) powers the ESP32 directly. No battery in the balaclava.

Architecture:

- Source: car 12V (ignition-switched)
- Conversion: DC-DC buck regulator 12V → 5V, 1A (LM2596 or MP1584)
- Cable: thin flexible PTFE 28 AWG, from dashboard to balaclava via helmet neck
- Protection: 500 mA fuse on the car side + 100μF decoupling capacitor on ESP32 side

Trade-offs:

- Unlimited autonomy for any session length
- No LiPo inside the helmet — zero thermal risk, zero regulatory constraint
- No BMS/charge management in firmware — maximum simplicity
- Cable can carry data (UART) for stand-side debugging
- Not usable away from the car (simulator, paddock warm-up)

Option B — Standalone LiPo

A flat LiPo cell integrated into the system, managed by a BMS IC.

Trade-offs:

- Fully autonomous — usable everywhere (simulator, paddock, warm-up)
- 1200 mAh at 400 mA = ~3h autonomy. With firmware sleep (~150 mA average) = ~8h
- LiPo in a hot helmet (40–50°C): thermal risk + up to ~20% capacity loss at 45°C
- Added complexity: BMS, boost converter, charge connector

Option C — Hybrid (recommended for final version)

Primary wired power from the car + LiPo backup managed by a PMIC. Automatic seamless switchover. This is the architecture used in professional systems (laptops, medical wearables).

Target architecture:

- Primary source: wired 5V from the car
- Backup source: 3.7V LiPo 400–600 mAh (small size, lower thermal risk)
- PMIC: BQ25895 (Texas Instruments) — charge/discharge management, automatic switchover, SOC monitoring via I²C
- Charge: USB-C when stored, or inductive (Qi) via a flat coil in the helmet support
- SOC data exposed to firmware → battery level widget on the React dashboard

INFO The FIA uses inductive charging for F1 biometric gloves (Signal Biometrics): the driver places the gloves on a charging mat, no manual handling required. This is the UX standard to target for a product version.

3. Comparison table

Criterion	A — Wired	B — LiPo	C — Hybrid
Autonomy	Unlimited ✓	~3h–8h ✓	4h+ backup ✓
HW complexity	Low ✓	Medium	High △
Thermal risk	None ✓	LiPo in helmet △	Small LiPo ✓
Off-car use	No x	Yes ✓	Yes ✓
Simulator / paddock	No x	Yes ✓	Yes ✓
Cost	Very low ✓	Medium	High
V1 reliability	Maximum ✓	Correct	Complex

4. Decision — V1 retained: Option B (LiPo standalone)

OK Option B retained for V1. LiPo 1200 mAh + TP4056+DW01A + MT3608 boost. Fully standalone system — usable on circuit, simulator, and paddock. All components already available.

Rationale: BalaSense is a personal project with no deadline pressure and all battery components already on hand. Option B adds ~2–3h of integration work and makes the system fully autonomous from day one — usable for simulator sessions, paddock warm-up, and bench testing without any car dependency. The thermal risk is managed by the TP4056+DW01A protection circuit and the flat LiPo format integrated into the HANS enclosure.

4.1 V1 hardware actions

Item	Specification
Battery	LiPo 1200 mAh flat format, JST-PH 2-pin connector
Charge IC	TP4056 + DW01A (2-chip module with protection) — verify 2 chips on module before ordering
Boost converter	MT3608, 3.7V → 5V, adjusted to 5V output — powers GSR (CJMCU-6701) and MAX30003 VIN
Decoupling capacitors	100 nF ceramic per sensor (x6) + 10 µF electrolytic on 3.3V and 5V rails
Voltage monitoring	Resistor divider on GPIO34 — monitors supply voltage, MQTT alert on drop

4.2 Decoupling capacitor placement

Each sensor receives a 100 nF ceramic capacitor soldered directly across its VCC and GND pins on the perfboard (5×7 cm). Placement priority:

- MAX30003 (ECG): highest priority. ECG is a low-amplitude signal; VIN noise appears directly in the waveform. Solder 100 nF ceramic + 10 µF electrolytic as close as possible to VIN/GND.
- SCD41, SDP810, TMP117, ICM-20948: 100 nF ceramic each, on the 3.3V rail.
- CJMCU-6701 (GSR): 100 nF ceramic on the 5V rail.
- Bulk: 10 µF electrolytic on the 3.3V rail output (ESP32) and on the Step-Up 5V output.

Parts: 100 nF ceramic 50V × 10 pcs + 10 µF electrolytic 25V × 5 pcs. Available as ~2€ assortment kits.

5. V2/V3 design decisions to anticipate now

5.1 Standardise the power connector

On V1 perfboard, provision a JST-PH 2-pin connector for the 5V input. This same connector will be reused in V2 (LiPo + boost) and V3 (PMIC). Standardising this entry point avoids rewiring later.

5.2 I²C address reservation for the PMIC

The V1 I²C bus currently occupies: SCD41 (0x62), ICM-20948 (0x68), SDP810 (0x25), TMP117 (0x48).

NOTE MAX77650 uses address 0x48 — direct conflict with TMP117. Prefer BQ25895 (0x6A) for V3 to avoid any I²C collision.

5.3 LiPo form factor for V2

Target flat cells (thickness < 7 mm) for integration in the HANS enclosure C3 compartment (26 × 70 × 22 mm internal). A 1200 mAh cell typically measures ~50 × 35 × 6 mm (e.g. format 603450). Verify exact dimensions against the chosen cell before finalising the CAD model.

5.4 Firmware power optimisation (V2)

- ESP32 modem sleep between MQTT publish bursts: peak 340 mA → target < 150 mA average
- SCD41 duty cycle: measurement every 5 s is already power-efficient; extend to 10 s if autonomy is critical
- MAX30003 FIFO depth: read every 250 ms instead of continuous polling